Data Structures

Mushtari Sadia

Introduction

Faculty Name: Mushtari Sadia (MIS)

Consultation Hours:

Wednesday: 9:30AM-1:45PM

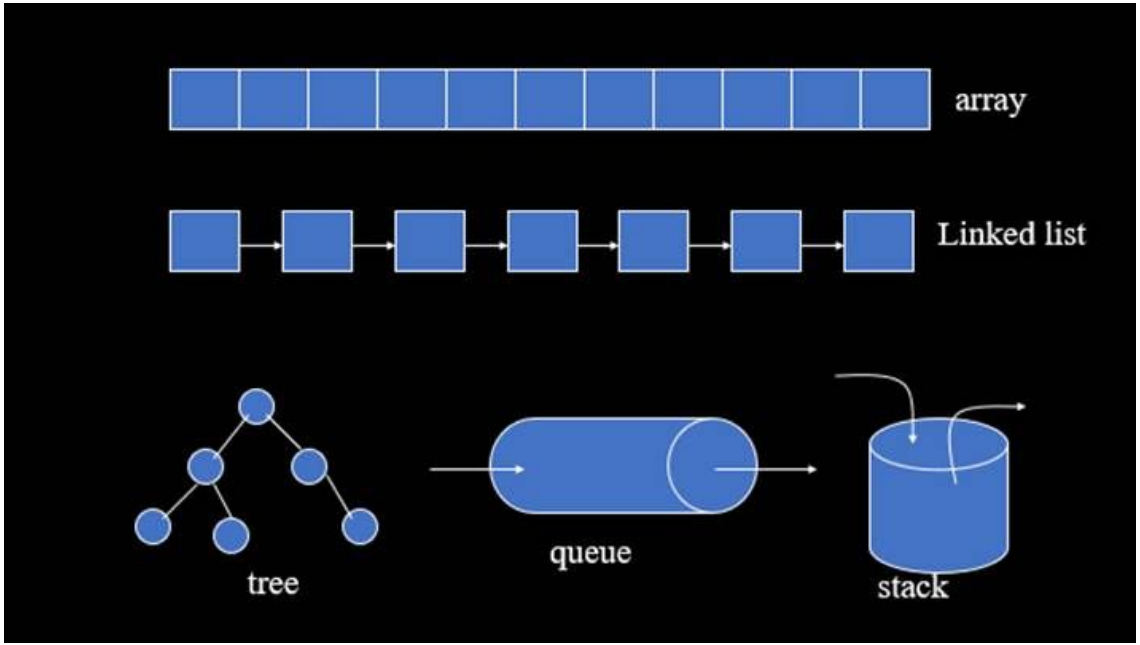
Thursday: 9:30AM-10:40AM

Room: **UB-80601**

Join Slack Channel:
https://join.slack.com/t/cse220summer2023/shared_invite/zt-1wrcesdkw-7bGTJMs62ihUWRIq4RVlAA
Resources link: tiny.cc/**cse220mis**

What is Data Structure?

Data structure is a format for **organizing** and **storing** data



Reference Books

Sl.	Title	Author(s)	Publication Year	Edition	Publisher	ISBN
1	Algorithms in Java	Robert Sedgewick and Kevin Wayne	2011	4th Edition	Addison-Wesley	ISBN-10: 032157351X ISBN-13: 9780321573513
2	Introduction to Algorithms	Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest and Clifford Stein	2009	3rd Edition	MIT Press	ISBN-10: 0262033844 ISBN-13: 9780262033848

CLRS Book link:
https://drive.google.com/file/d/1JUrGXFldwjUsRRLbo0Ac6x9scMpe9Ayd/view?usp=sharing

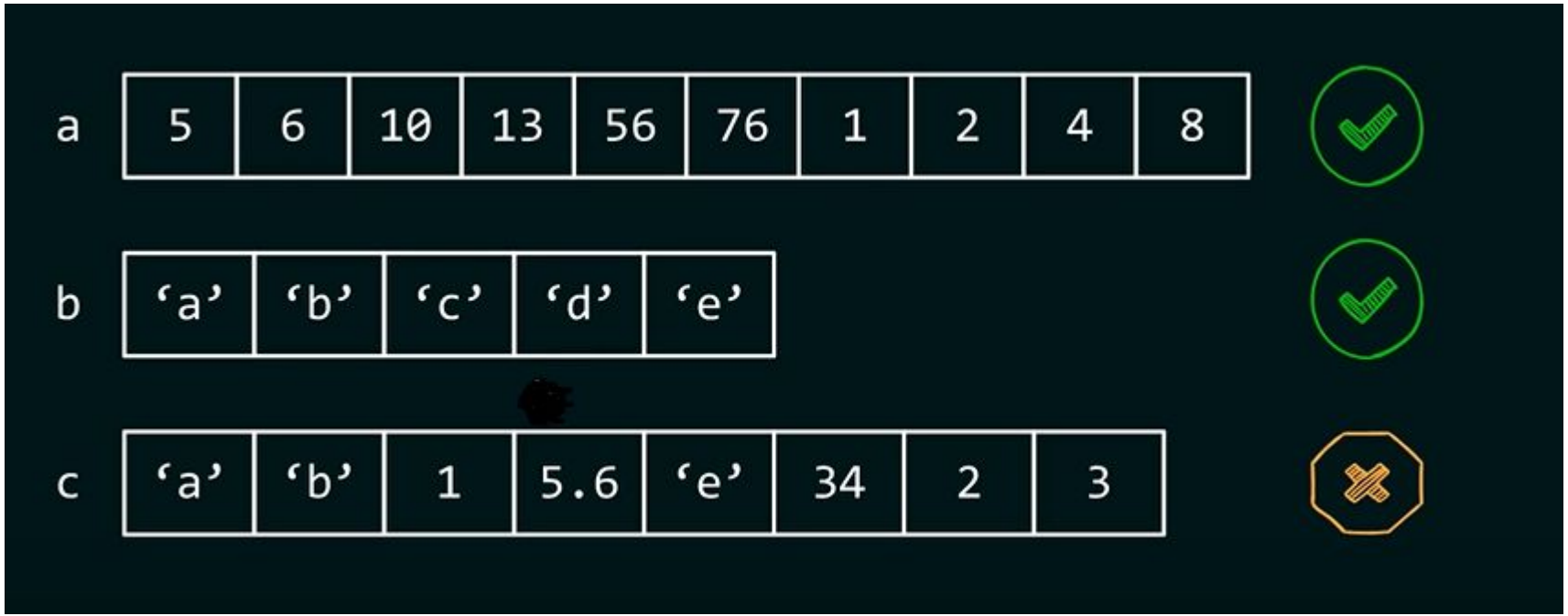
Linear Arrays and Circular Arrays

Week 1

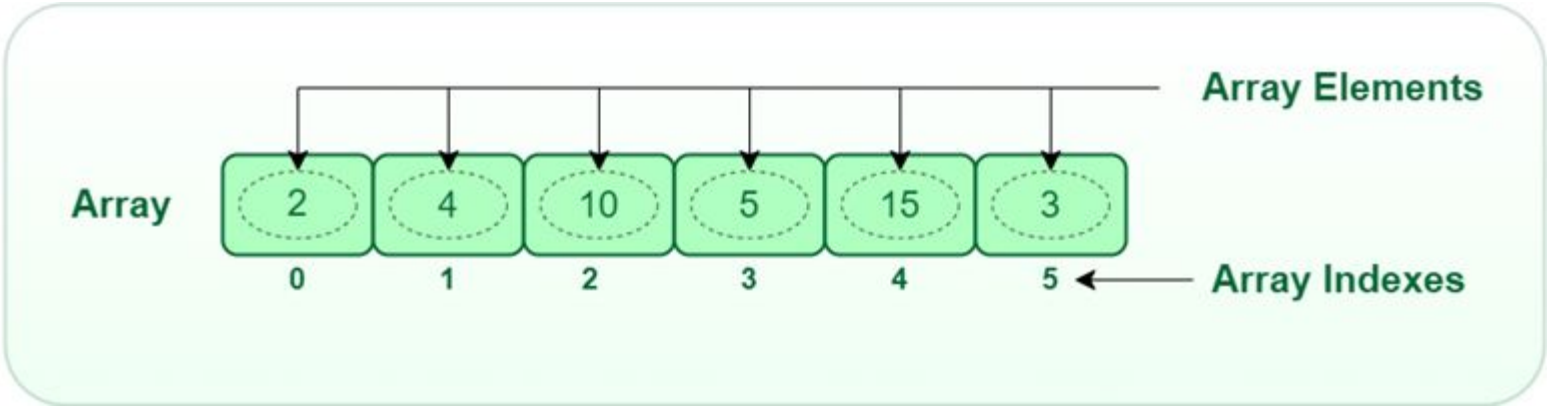
Outline

1. Definition of an Array
2. Properties of an Array
3. Operations on an Array

Linear Array



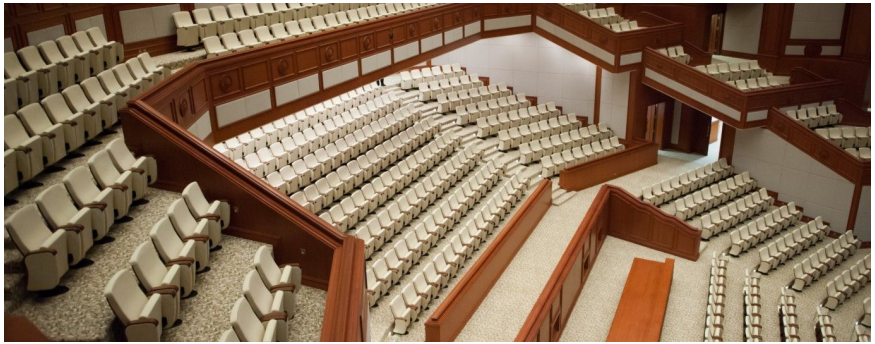
Linear Array



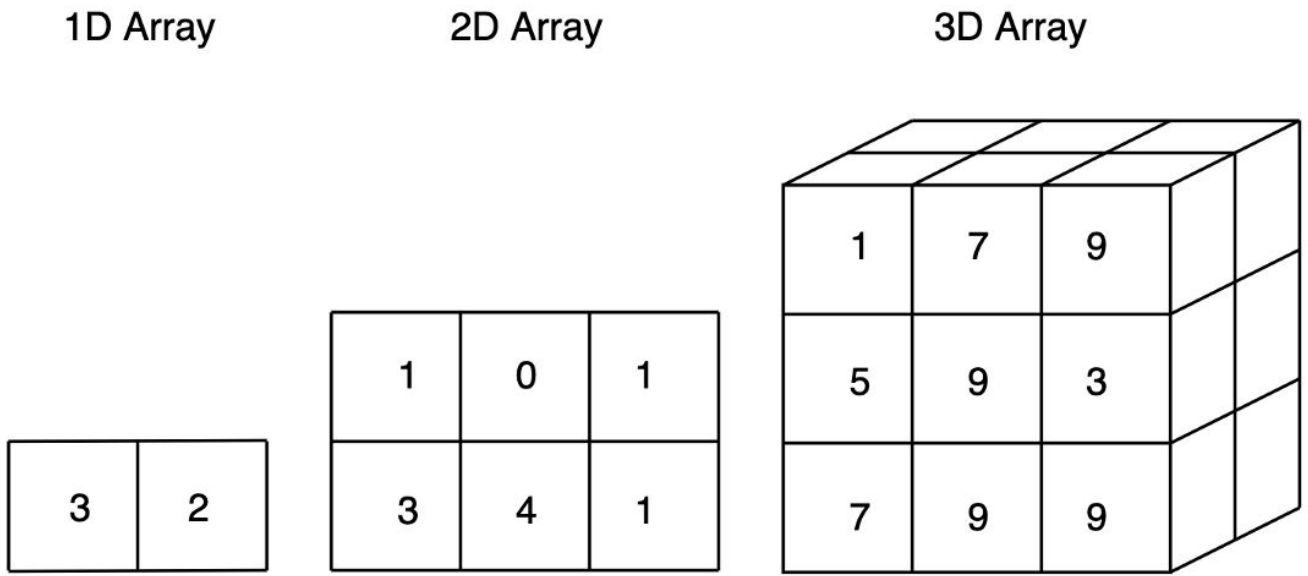
Linear Array - Properties

1. Dimension
2. Index
3. Length
4. Single Type

Dimension



Dimension



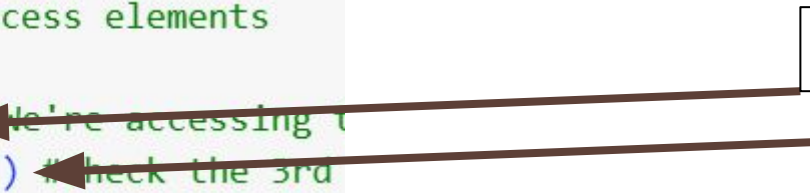
Dimension

```
#Let's create a one-dimensional array of five elements
array_1D = [None]*5
```

```
#Let's create a two-dimensional array of size 4x5
array_2D = [None]*4
for i in range(len(array_2D)):
    array_2D[i] = [None]*5
```

```
#Let's see how to access elements

print(array_1D[3]) #We're accessing t
print(array_2D[2][3]) #Check the 3rd
```



Index

```
#Let's create a one-dimensional array of five elements
array_1D = [None]*5
```

```
#Let's create a two-dimensional array of size 4x5
array_2D = [None]*4
for i in range(len(array_2D)):
    array_2D[i] = [None]*5
```

```
#Let's see how to access elements

print(array_1D[3]) #We're accessing t
print(array_2D[2][3]) #Check the 3rd
```

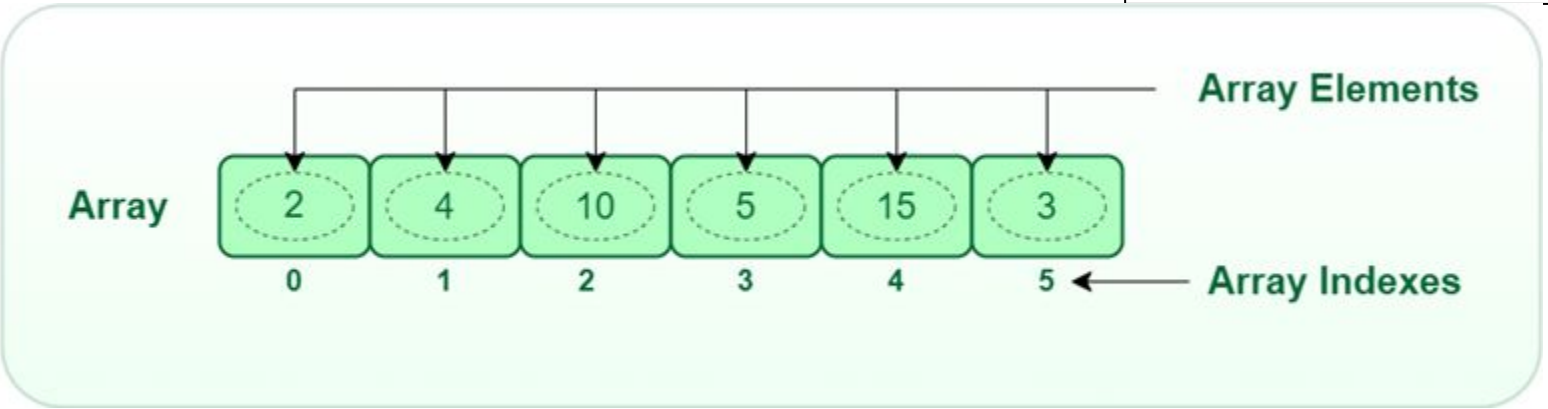
Index of an array starts from 0;
So from now on, always start counting from 0!

3rd element of the array

3rd element of 2nd row

Index

Index of an array starts from 0;
So from now on, always start counting from 0!



Length

When you create an array in the program, you have to specify the length of each of its dimensions that **you cannot change later**

Length

```
# Let's look at the length property clearly
array = [None]*10
#We just declared an array of 10 length. Let's fill it up with something.

Names = ['Emon','Nusrat','Joy','Abrar','Ifty','Ashik','Raisa','Suba','Sazid','Jawad']
for i in range(10):
    array[i] = Names[i]
print(array)
```

```
['Emon', 'Nusrat', 'Joy', 'Abrar', 'Ifty', 'Ashik', 'Raisa', 'Suba', 'Sazid', 'Jawad']
```

Length

```
[ ]  1  print(array[1])
```

```
Nusrat
```

```
[13]  1  print(array[11])
```

```
---------------------------------------------------------------------------
IndexError                                Traceback (most recent call last)
<ipython-input-13-92562eb7d277> in <cell line: 1>()
----> 1 print(array[11])

IndexError: list index out of range
```

SEARCH STACK OVERFLOW

```
[14]  1  print(array[0])
```

```
Emon
```

```
  1  print(array[10])
```

```
---------------------------------------------------------------------------
IndexError                                Traceback (most recent call last)
<ipython-input-15-a0db80d8cb8e> in <cell line: 1>()
----> 1 print(array[10])
```

Operations On Array - Read and Write

Operations on Array

The most common operations that you do on an array are writing/reading a data item to/from an index. Suppose you have a 1D array named *student_names*. If you want get the 5th student from the array then you write:

fifth_student = student_names[5]

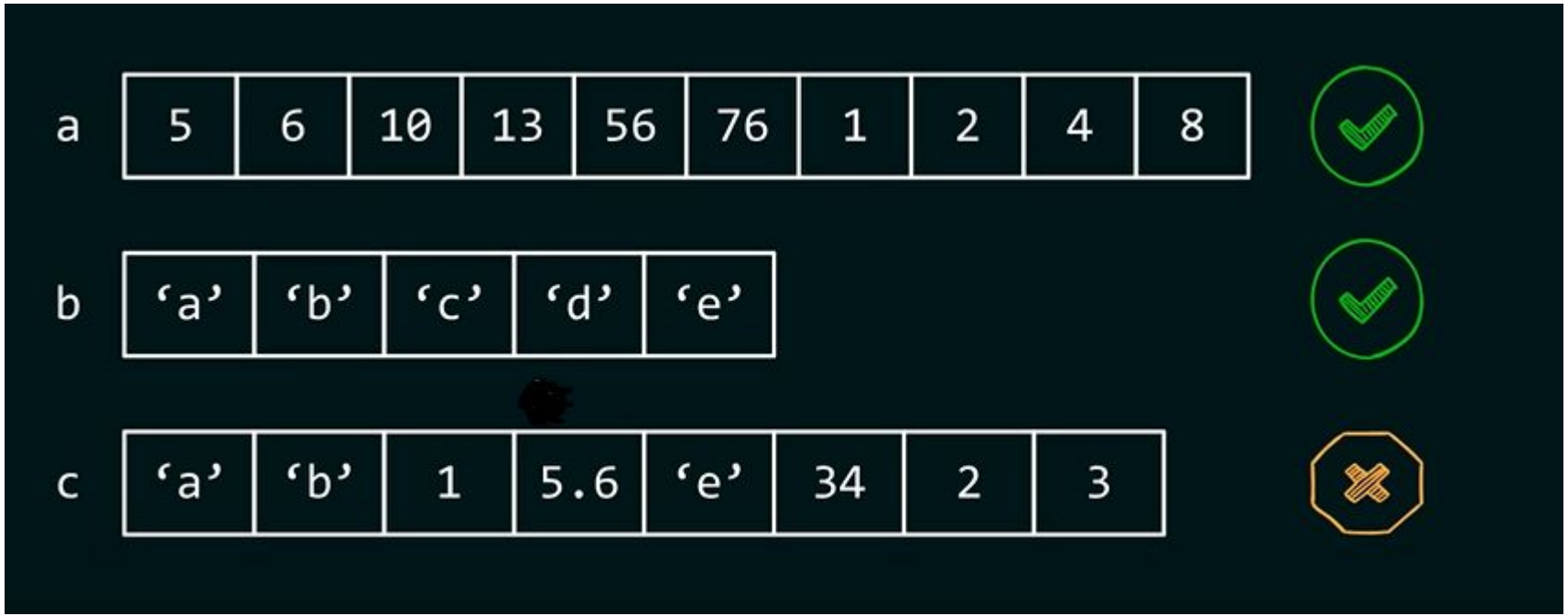
To change the value at 5th index and set it to 'John Doe', you write:

student_names[5] = 'John Doe'

Single Type

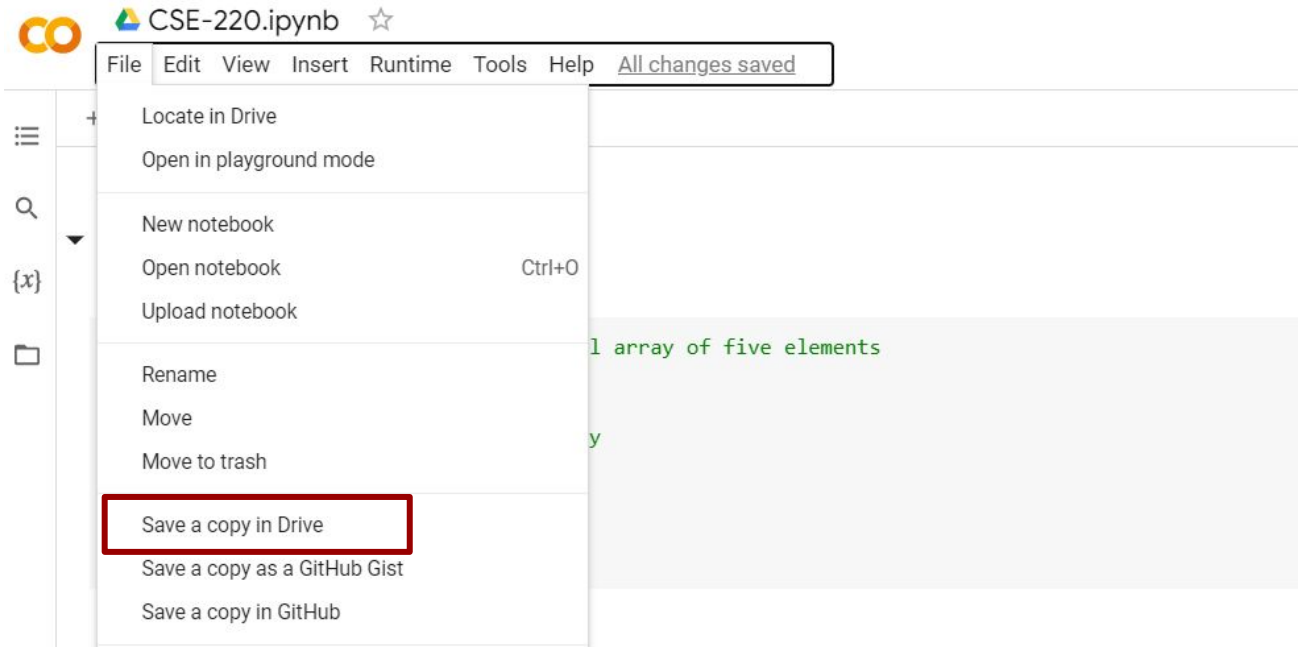
- An array can hold elements of only a single type. That is, you cannot mix strings with numbers in an array – not even mix integer numbers with fractional numbers.

Single Type



Practice

Notebook Link: http://tiny.cc/cse220nb



Some other functions

Iteration: Iteration refers to checking all the values index by index. The main idea is to go to that memory location and check all the values index by index.

```
1 def iteration(source):
2   for i in range(len(source)):
3     print(source[i])
4
5 def reverseIteration(source):
6   for i in range(len(source) - 1, -1, -1):
7     print(source[i])
```

Some other functions

Copy Array: Copy array means you initialize a new array with the same length as the given array to copy and then copy the old array's value by value. As the variable where we store the array only stores the memory location, only copying the value is not enough for array copying. For example, if you have an array titled arr = [1, 2, 3, 4] and write a2 = arr. It does not mean you have copied arr to a2.

```
1 def copyArray(source):
2     newArray = [None] * len(source)
3     for i in range(len(source)):
4         newArray[i] = source[i]
5     return newArray
```

Some other functions

Resize Array: We can not resize an array because it has a fixed memory location. However, if we ever need to resize an array, we need to create a new array with a new length and then copy the values from the original array. For example, if we have an array [10, 20, 30, 40, 50] whose length is 5 and want to resize the array with length 8. The new array will be [10, 20, 30, 40, 50, None, None, None].

```
1 def resizeArray(oldArray, newCapacity):
2   newArray = [None] * newCapacity
3   for i in range(len(oldArray)):
4     newArray[i] = oldArray[i]
5   return newArray
```

Some other functions

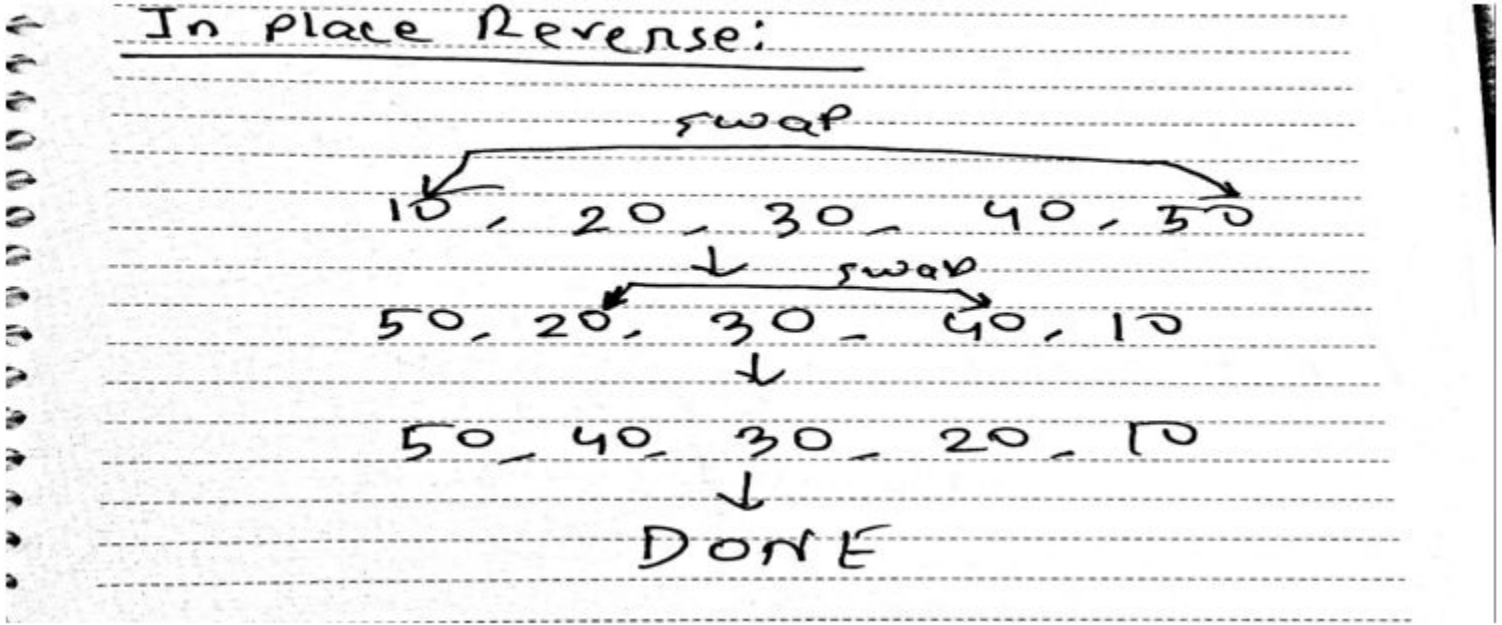
Reversing an Array: Reversing an array can be implemented in two ways. First, we will create a new array with the same size as the original array and then copy the values in reverse order. The method is called out-of-the-place operation.

```
1 def reverseArrayOutOfPlace(arr):
2   revArr = [None] * len(arr)
3   i = 0
4   j = len(arr) - 1
5   while i < len(arr):
6     revArr[i] = arr[j]
7     i += 1
8     j -= 1
9   return revArr
```

Some other functions

However, an efficient approach might be to reverse the array in the original array. By this, we will not need to allocate extra spaces. This is known as an in-place operation. To do so we need to start swapping values from the beginning position to the end position. The idea is to swap starting value with the end value, then the second value with the second last value, and so on.

Some other functions



Some other functions

```
1 def revArrInPlace(arr):
2     i = 0
3     j = len(arr) - 1
4     while i < j:
5         temp = arr[i]
6         arr[i] = arr[j]
7         arr[j] = temp
8         i += 1
9         j -= 1
```

Functions on array

Shifting an Array Left: Shifting an entire array left moves each element one (or more, depending how the shift amount) position to the left. Obviously, the first element in the array will fall off at the beginning and be lost forever. The last slot of the array before the shift (ie. the slot where the last element was until the shift) is now unused (we can put a None there to signify that). The size of the array remains the same however because the assumption is that you would something in the now-unused slot. For example, shifting the array [5, 3, 9, 13, 2] left by one position will result in the array [3, 9, 13, 2, None]. Note how the array[0] element with the value of 5 is now lost, and there is an empty slot at the end.

Functions on array

```
1 def shiftLeft(arr):
2     for i in range(1, len(arr)):
3         arr[i-1] = arr[i]
4     arr[len(arr) - 1] = None
5     return arr
```

Functions on array

Shifting an Array Right: Shifting an entire array right moves each element one (or more, depending how the shift amount) position to the right. Obviously, the last element in the array will fall off at the end and be lost forever. The first slot of the array before the shift (ie., the slot where the first element was until the shift) is now unused (we can put a None there to signify that). The size of the array remains the same however because the assumption is that you would something in the now-unused slot. For example, shifting the array [5, 3, 9, 13, 2] right by one position will result in the array [None, 5, 3, 9, 13]. Note how the array[4] element with the value of 2 is now lost, and there is an empty slot at the beginning.

Functions on array

```
1 def shiftRight(arr):
2   for i in range(len(arr) - 1, 0, -1):
3     arr[i] = arr[i - 1]
4   arr[0] = None
5   return arr
```

Functions on array

Rotating an Array Left: Rotating an array left is equivalent to shifting a circular or cyclic array left where the 1st element will not be lost, but rather move to the last slot. Rotating the array [5, 3, 9, 13, 2] left by one position will result in the array [3, 9, 13, 2, 5].

Functions on array

```
1 def rotateLeft(arr):
2     temp = arr[0]
3     for i in range(1, len(arr)):
4         arr[i-1] = arr[i]
5     arr[len(arr) - 1] = temp
6     return arr
```

Functions on array

Rotating an Array Right: Rotating an array right is equivalent to shifting a circular or cyclic array right where the last element will not be lost, but rather move to the 1st slot. Rotating the array [5, 3, 9, 13, 2] right by one position will result in the array [2, 5, 3, 9, 13].

Functions on array

```
1 def rotateRight(arr):
2   temp = arr[len(arr) - 1]
3   for i in range(len(arr) - 1, 0, -1):
4     arr[i] = arr[i - 1]
5   arr[0] = temp
6   return arr
```

Functions on array

Inserting an element into an Array: To insert an element into the array we need to make an empty slot first and then insert the value in the array. To make an empty slot first we need to check if any slots are available or not. If slots are not available then we need to resize the array using the concept of array resizing mentioned above. After that, we will use the idea of the right shift so that we can have an empty slot. The initialization point of the right shift would be the index where we want to insert the value. After the right shift operation, we will insert the value in that corresponding index.

For example, we have an array consisting of values [5, 7, 3, 6, None, None] (None represents the empty slots). Now if we want to insert 10 at index 1 we will need to use the right shift starting from index 1. So our array will be like [5, None, 7, 3, 6, None]. Here we can use the index 1 and insert the value 10 and the final array would be [5, 10, 7, 3, 6, None]

Functions on array

```
1 def insertElement(arr, size, elem, index):
2   # Practice how to throw exception if there is no empty space
3   if size == len(arr):
4     print("No space left. Insertion failed")
5   else:
6     for i in range(size, index, -1):
7       arr[i] = arr[i - 1] #Shifting right till the index
8     arr[index] = elem #Inserting element
9     return arr
```

Functions on array

Removing an element from the Array: We need to use the concept of left shift to remove an element from the array. The idea is to start the left shift from the index which value we want to remove.

For example, we have an array consisting of values [10, 6, 8, 11, 15, 20] and want to remove value 8 which is at index 2, we need to do the left shift operation starting from index 2. So the resulting array would be [10, 6, 11, 15, 20, None].

Functions on array

```
1 def removeElement(arr, index, size):
2   for i in range(index + 1, size):
3     arr[i - 1] = arr[i] #Shifting left from removing index
4   arr[size - 1] = None #Making last space empty
```

How are multi-dimensional arrays stored?

The following diagram shows the difference between these two approaches for a 2D matrix of 3×3 dimensions. (By the way, row is the vertical and column is the horizontal axis in 2D. You should already know that. Here Dimension 0 represents the row and Dimension 1 represents the column.)

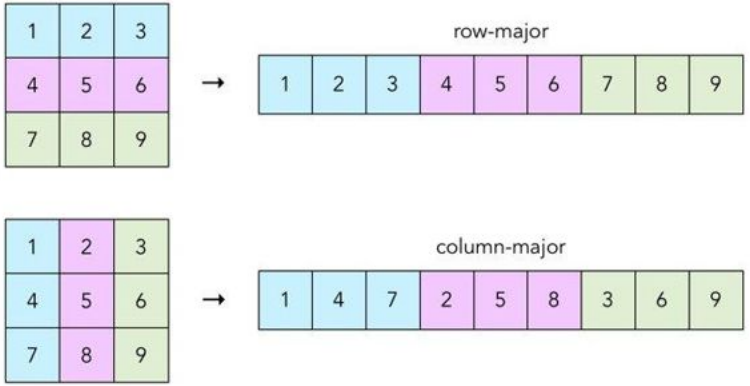


Figure 1: Two Different Ways of Storing Multidimensional Arrays in a Linear Format

How are multi-dimensional arrays stored?

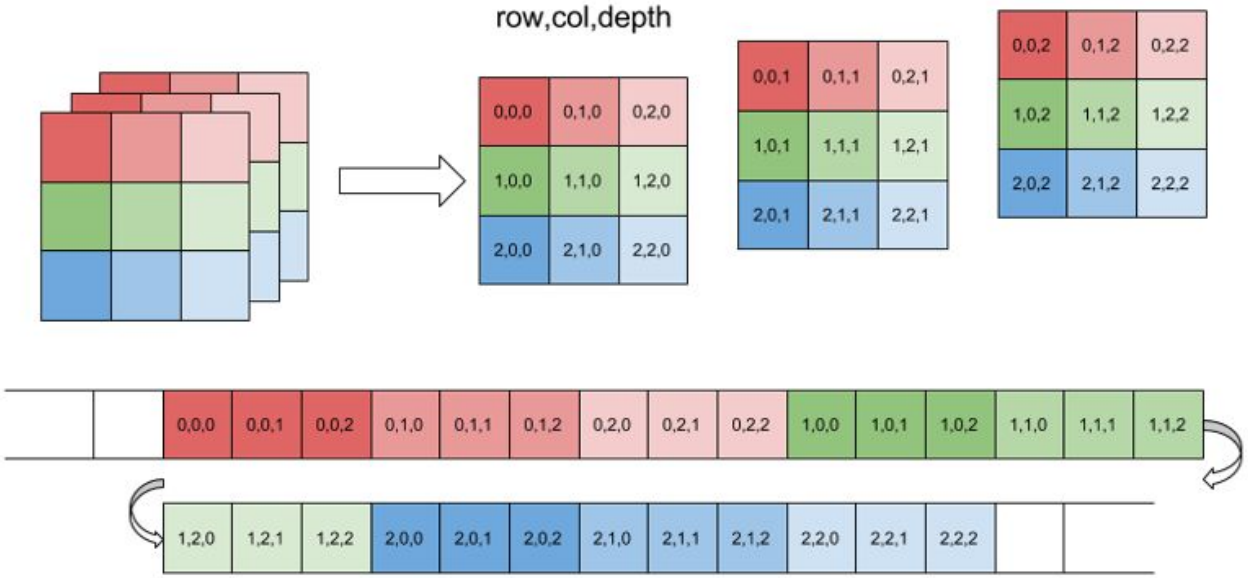


Figure 2: Row Major Ordering of Indexes of a 3D Array

Multidimensional Array Index -› Linear Index

imagine you have a 4D array with dimensions M×N×O×P, named *box*

element at the index you are interested is the *box[w][x][y][z]*

Linear Index = $w \times (N \times O \times P) + x \times (O \times P) + y \times P + z$

Multidimensional Array Index -› Linear Index

imagine you have a 4D array with dimensions M×N×O×P, named *box*

element at the index you are interested is the *box[w][x][y][z]*

Linear Index = $w \times (N \times O \times P) + x \times (O \times P) + y \times P + z$

Multidimensional Array Index -› Linear Index

imagine you have a 4D array with dimensions M×N×O×P, named *box*

element at the index you are interested is the *box[w][x][y][z]*

Linear Index = $w \times (N \times O \times P) + x \times (O \times P) + y \times P + z$

Multidimensional Array Index -› Linear Index

imagine you have a 4D array with dimensions M×N×O×P, named *box*

element at the index you are interested is the *box[w][x][y][z]*

Linear Index = $w \times (N \times O \times P) + x \times (O \times P) + y \times P + z$

Multidimensional Array Index -› Linear Index

imagine you have a 4D array with dimensions M×N×O×P, named *box*

element at the index you are interested is the *box[w][x][y][z]*

Linear Index = $w \times (N \times O \times P) + x \times (O \times P) + y \times P + z$

Multidimensional Array Index -› Linear Index

imagine you have a 4D array with dimensions 4x3x5x6, named *box*

element at the index you are interested is the box[2][1][4][3]

Linear Index = ??

Linear Index -› Multidimensional Array Index

Imagine you have a 3D array with dimensions 4x4x8.
What are the multidimensional indexes of the element stored at location 111?

Linear Index = 111 -> box[x][y][z]

Find x,y,z

Linear Index -› Multidimensional Array Index

Imagine you have a 3D array with dimensions 4x4x8.
What are the multidimensional indexes of the element stored at location 111?

Linear Index = x * (4*8) + y*8 + z

= **111**

Linear Index -› Multidimensional Array Index

Imagine you have a 3D array with dimensions 4x4x8.
What are the multidimensional indexes of the element stored at location 111?

Linear Index = x * (4*8) + y*8 + z

where 0 <= x <= 3, 0 <= y <= 3 and 0 <= z <= 7

= 111

Linear Index -› Multidimensional Array Index

Imagine you have a 3D array with dimensions 4x4x8.
What are the multidimensional indexes of the element stored at location 111?

Linear Index = x * (4*8) + y*8 + z

= 111

where 0 <= x <= 3, 0 <= y <= 3 and 0 <= z <= 7

Remember that indexing starts from 0 in each dimension!

Linear Index -› Multidimensional Array Index

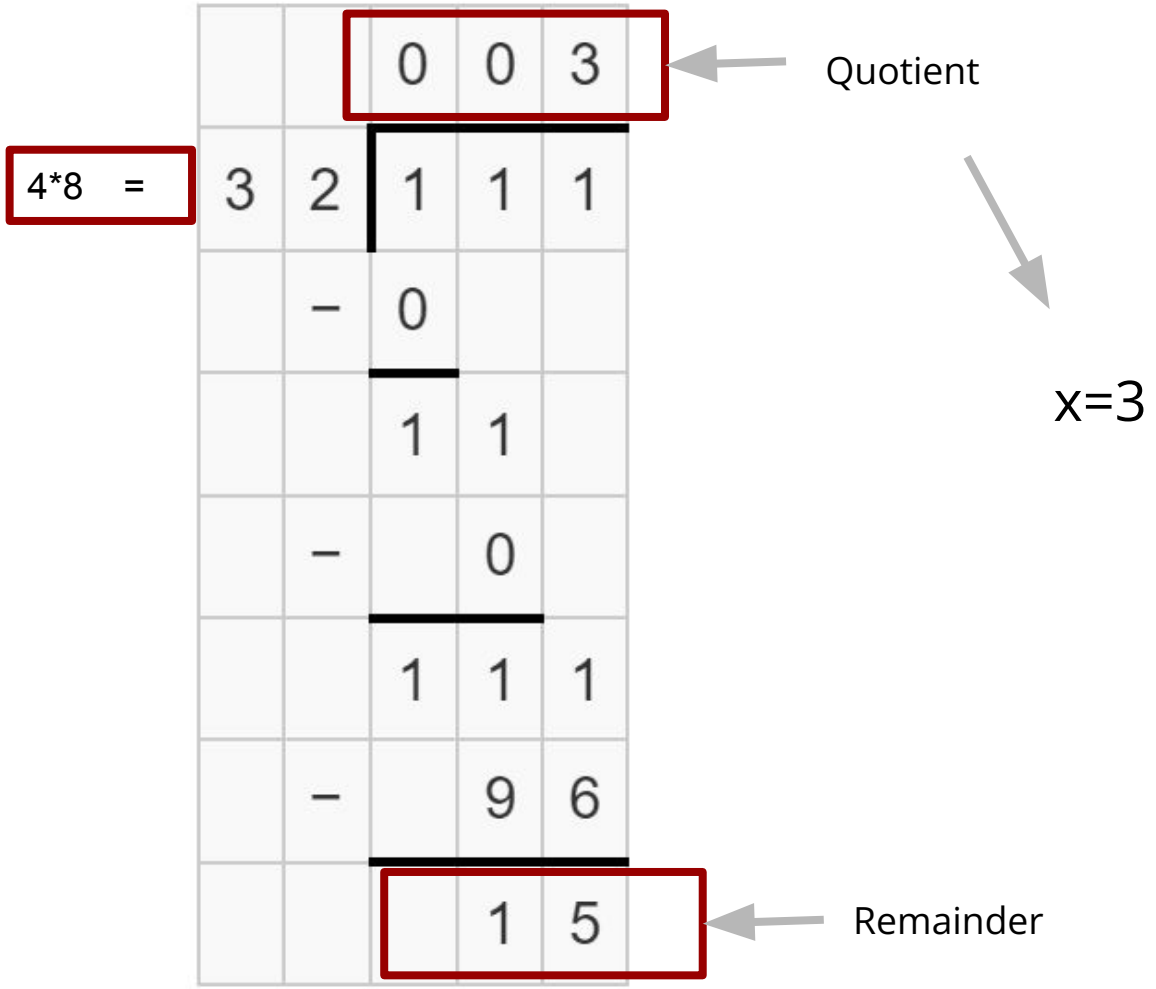
Imagine you have a 3D array with dimensions 4x4x8.
What are the multidimensional indexes of the element stored at location 111?

Linear Index = x * (4*8) + y*8 + z

= **111**

where 0 <= x <= 3, 0 <= y <= 3 and 0 <= z <= 7

Remember that indexing starts from 0 in each dimension!


0 0 3
Quotient
4*8 =
3 2 1 1 1
− 0
1 1
− 0
1 1 1
− 9 6
1 5
Remainder
x=3

Linear Index -› Multidimensional Array Index

Following the process we get,

X = 111//(4*8) = 3 and 111 % (4*8) = 15

Y = 15 // 8 = 1 and 15 % 8 = 7

Z = 7

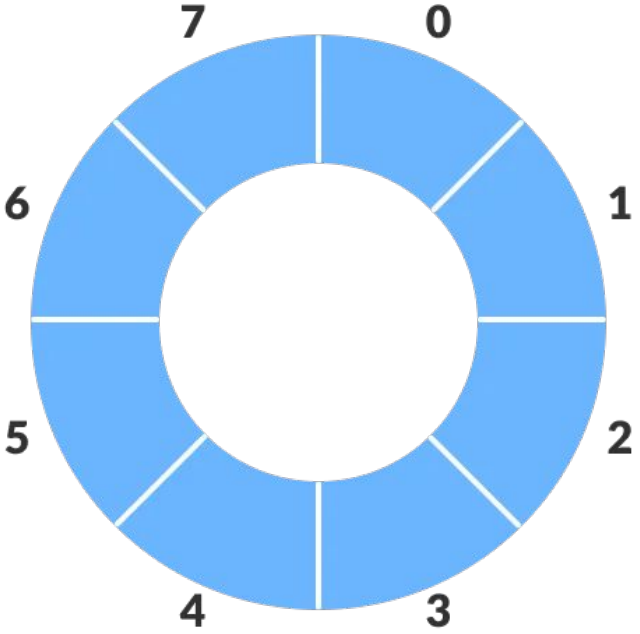
So, the dimension of linear index 111 is [3][1][7].

Linear Index -› Multidimensional Array Index

Practice

Find the dimension of 96, 107, and 60 of the linear index.

Circular Array/Buffer



Indexing A Circular Array

Assume the capacity/length of a circular array, named buffer, is 10.

Suppose, you want to read the element at index 9.

a = buffer[9 % 10]

= buffer[9]

Indexing A Circular Array

Assume the capacity/length of a circular array, named buffer, is 10.

What if you ask what is the next index after 9?

a = buffer[(9 + 1) % 10]

= buffer[0]

Indexing A Circular Array

This works for going **leftwards** also.

For example, if you ask for the element that is two steps left of the element at index zero; then you do the following:

a = buffer[(0 - 2) %10]

The answer is (0-2)%10 = -2%10 = 8

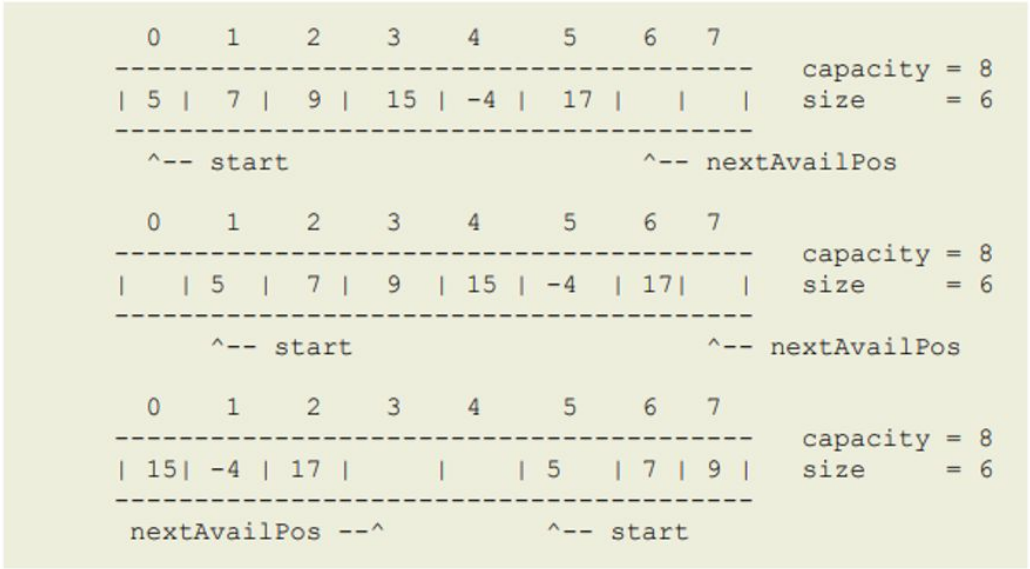
Circular Array Structure

```
  0    1    2    3    4    5    6    7
-----------------------------------------   capacity = 8
| 5 |  7 |  9 |  15 | -4 |  17 |    |    |   size     = 6
-----------------------------------------
                                  ^-- nextAvailPos = size
```

Here capacity represents the fixed array length and size represents the number of elements present in the array at that time. For example, if 17 was not present then the capacity would still be the same which is 8 but the size would be 5.

Circular Array Structure

Now check the following image where this linear array is converted to a circular array with different start indexes.



Iteration Over A Circular Array

Forward:

```
1 # Forward Iteration
2 def forwardIteration(cir, start, size):
3     k = start
4     for i in range(size):
5         print(cir[k])
6         k = (k + 1) % len(cir)
7
```

Iteration Over A Circular Array

Backward:

Last item's index of circular array = (start + size - 1) % len(cir). You will understand the formula when you read the offset part.

```
 8 # Backward Iteration
 9 def backwardIteration(cir, start, size):
10     k = (start + size - 1) % len(cir)
11     for i in range(size):
12         print(cir[k])
13         k = (k - 1) % len(cir)
```

Self Study